

Laboratory 8:

Matrix-vector optimization

Objective:

- memory access optimization in a simple matrix-vector procedure

Tasks:

- 1) Create a working directory (eg. lab_8) .
- 2) Copy and extract [files](#).
- 3) Execute the my_program and measure the time of execution for different compiler optimization options (eg. O0 and O3)
- 4) Obtain the assembler code files for different levels of compiler optimization (eg. O0 and O3)
 - gcc -S -O0 mat_vec.c
 - cp mat_vec.s mat_vec.gcc_O0
- 5) Analysis and the comparison of the obtained assembler codes:
 - a) Select the code snippets corresponding to each instruction in C (you can use <https://godbolt.org/> after changing the options to get the same code as on your local computer).
 - b) Count the number of memory accesses in the code.
- 6) Create a new procedure mat_vec_1 with a more optimal access, execute, measure the time and analyze the assembler for the fastest code obtained.
- 7) Measure the performance of the worst and best code and compare it with the theoretical values.

Assessment:

- Class attendance and a report with all results (time comparison!) and assembler analysis

Additional information:

- [Intel](#),
- [AMD](#),
- [Short](#)